



Harish Kumar S 251801090-2025-AIDS ▾

**H2****Started on** Thursday, 24 September 2026, 8:21 AM**State** Finished**Completed on** Thursday, 24 September 2026, 8:26 AM**Time taken** 5 mins 43 secs**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct | Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

**Example 1:****Input:** 6**Output:** 6**Explanation:** There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

**Input Format**

First Line contains the number n

**Output Format****Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      long long dp[n + 1];
8
9      dp[0] = 1;
10
11     for (int i = 1; i <= n; i++) {
12         dp[i] = dp[i - 1];
13
14         if (i >= 3) {
15             dp[i] += dp[i - 3];
16         }
17     }
18
19     printf("%lld", dp[n]);
20
21     return 0;
22 }
23

```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 6     | 6        | 6   | ✓ |

|   | Input | Expected          | Got               |   |
|---|-------|-------------------|-------------------|---|
| ✓ | 25    | 8641              | 8641              | ✓ |
| ✓ | 100   | 24382819596721629 | 24382819596721629 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

[Back to Course](#)